

Deanonimisation of Encrypted Communications

Background

With the rapid growth of cyber-enabled crimes, organized criminal groups increasingly rely on **end-to-end encrypted and anonymized communication platforms** to avoid detection. Applications and privacy-focused communication channels conceal message content, user identity, and routing information, making traditional digital forensic investigations ineffective.

Conventional device forensics often fails to establish communication linkage, timing correlation, or network attribution when encrypted messaging services are used. As a result, investigators face significant challenges in identifying remote communicating entities involved in suspicious activities.

Challenge Overview

Participants are required to design and develop a **lawful and non-intrusive forensic tool** capable of analyzing **network-level metadata** from a suspect's device.

The proposed solution should focus on:

- Capturing communication-related network traffic
- Analyzing traffic patterns without decrypting data
- Correlating metadata to identify communication behavior
- Assisting investigators in narrowing down or characterizing the remote communicating party (Party B)

Decryption of message content is NOT required or expected.

Objectives

The solution should aim to:

- Monitor encrypted communication traffic legally
- Perform packet capture and metadata extraction

- Identify communication frequency and timing patterns
- Detect possible communication endpoints
- Provide behavioral or attribution insights using observable network activity

Expected Solution Features

Participants may include (but are not limited to):

- Network packet capture mechanisms
- Traffic flow and session analysis
- Metadata correlation and visualization
- Communication timing analysis
- IP/network behavior profiling
- Dashboard for investigative insights

No message decryption

No unauthorized surveillance techniques